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Prof. Jin Xuan and Prof. Jinfeng Liu
Editors-in-Chief
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Dear Professors Xuan and Liu,

Please consider my original research article, “Optimization of a Recycling N-Stage Activated Sludge System with a Clarifier using A Physically Constrained Surrogate,” for publication in *Digital Chemical Engineering*.

The manuscript develops a closed-loop design-and-operation framework for a prescribed N -stage activated sludge system connected to a secondary Clarifier. The number of stages is a fixed topology parameter, and the case study uses five continuous stirred-tank reactors. The decisions include nominal hydraulic retention time, zonal aeration, mixed-liquor recycle, return activated sludge, and waste activated sludge. One dimensionless second-order surrogate jointly predicts the mixer, every reactor outlet, Clarifier outlet component mass flows, and the ten-layer Clarifier solids profile, giving 170 response coordinates in the case study.

The physical deployment calculation is a unique scaled- L_2 lower-level quadratic program for the interconnected bioreactor–Clarifier network. It enforces exact recycle mixing, interstage stoichiometric invariants, componentwise Clarifier mass balance, soluble pass-through, particulate clarification and densification directions, layer-endpoint and inventory consistency, and non-negativity. RAS densification is represented as particulate recovery into a smaller underflow rather than artificial solids generation. The upper level evaluates clarified effluent quality together with volume, aeration, recycle-pumping, and wasted-solids terms, while separate capacity and trust screens prevent compliance from being manufactured by the physical correction.

The study builds on the component-space and invariant ideas in Selerio Jr. (2026), *An interpretable statistical surrogate of activated sludge systems that preserves mass conservation and component non-negativity*, *Digital Chemical Engineering*, 20, 100329. The estimator uses 351 dimensionless, uniquely serialized quadratic features in one full-column-rank standardized multiresponse ordinary least-squares problem. Column-pivoted QR computes the fixed coefficient matrix, and an independent singular-value decomposition verifies rank, conditioning, and coefficient agreement.

Training and validation use a coupled ASM2d-TSN–multilayer-Clarifier model. A fixed 16,000-state development fit must pass an assessment on 4,000 states held out from that fit before the unchanged estimator is refitted on all 20,000 states for production use. Predictive fidelity is then evaluated at surrogate-selected decisions. Decision quality is assessed separately through an independent mechanistic search and reported as a gap to the best finite-budget incumbent found under the same bounds. The formulation and computational record distinguish satisfaction of the declared linear physical contract from evidence about the associated engineering decision.

The work is aligned with the journal’s scope in digital process engineering, mathematical programming, physically constrained surrogate modeling, and computational wastewater-process operation. The manuscript is original, has not been published previously, and is not under consideration elsewhere. I have approved its submission and declare no competing interests. Thank you for your

consideration.

Sincerely,

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